

Perfluorooctanesulfonate and related fluorochemicals in human blood samples from China.

Perfluorooctanesulfonylfluoride (POSF)-based compounds have been manufactured and used in a variety of industrial applications. These compounds degrade to perfluorooctanesulfonate (PFOS) which is regarded as a persistent end-stage metabolite and is found to accumulate in tissues of humans and wildlife. PFOS, perfluorohexanesulfonate (PFHxS), perfluorooctanoate (PFOA), and perfluorooctanesulfonamide (PFOSA) have been found in human sera from the United States. In this study, concentrations of PFHxS, perfluorobutanesulfonate (PFBS), PFOS, perfluorohexanoic acid (PFHxA), PFOA, perfluorononanoic acid (PFNA), perfluorodecanoic acid (PFDA), perfluoroundecanoic acid (PFUnDA), perfluorododecanoic acid (PFDoDA), and PFOSA were measured in 85 samples of whole human blood collected from nine cities (eight provinces) in China, including Shenyang (Liaoning), Beijing (Hebei), Zhengzhou (Henan), Jintan (Jiangsu), Wuhan (Hubei), Zhoushan (Zhejiang), Guiyang (Guizhou), Xiamen (Fujian), and Fuzhou (Fujian). Among the 10 perfluorinated compounds (PFCs) measured, PFOS was the predominant compound. The mean concentration of PFOS was greatest in samples collected from Shenyang (79.2 ng/mL) and least in samples from Jintan (3.72 ng/mL). PFHxS was the next most abundant perfluorochemical in the samples. No age-related differences in the concentrations of PFOA, PFOS, PFOSA, and PFHxS were observed. Gender-related differences were found, with males higher for PFOS and PFHxS, and females higher in PFUnDA. Concentrations of PFHxS were positively correlated with those of PFOS, while concentrations of PFNA, PFDA, and PFUnDA were positively correlated with those of PFOA. There were differences in the concentration profiles (percentage composition) of various PFCs in the samples among the nine cities.